Project 1

1. How did you approach the project?
    a. I followed what was in the class demonstration and break it down into 3 main components. First I made the Logger and test its output, then I move on to Encryption and implement its logic using Vigenère cypher, test its function and make sure it is correct. Then moving on the hard part which is the Driver where I connect them together. After finish with its logic, I implement a somewhat better UI for user to use it easier.
2. How did you organize the project? Why?
    a. I used java as the language for this project. First, I created Logger.java where it manages logging of the program activities. Then, I created Encryption.java to handles the encryption and decryption logic. Finally, I created Driver.java, this is the main interface for user interaction and coordination between each component.
3. What problems did you encounter?
    a. Logging Format: The log entries had duplicate timestamps and lacked proper formatting.
    b. History: The history was recording duplicate entries and display them in mixed.
    c. Password Setting: The log was showing [ERROR] even though it was successfully set the password.
    d. Encrypt/Decrypt Error Handling: The program was adding failed encryption/decryption attempts to history.
    e. UI: The menu wasn’t formatted the way I wanted and lacked clear instructions.
    f. History Display: It only show the history recorded without option to ”(Go Back)”.
    g. QUIT message: The program wasn’t loggin the QUIT message after exited.

h. Case Handling: Inconsistent handling of both uppercase and lowercase inputs.
i. Error Message: Some error messages wasn't being properly logged in the Logger.java file.
j. Redundant History: The program stored both input and output of successfully operations to the history.
k. Log File Format: Log file entries wasn't consistently formatted, making it hard to read.

4. How did you fix them?
   a. Logging Format: I fixed them by modified the log method in Driver.java to remove duplicate timestamps and format log entries.
   b. History: I replace ArrayList with LinkedHashSet to automatically remove the duplicates and implemented conversion of all history entries to uppercase for consistency.
   c. Password Setting: I updated setPasskey method in Encryption.java to return the correct [SUCCESS] message and modified Driver.java to correctly interpret and log the password set response.
   d. Encrypt/Decrypt Error Handling: I modified the handleEncryptDecrypt method to only add successful operations to the history.
   e. UI: I redesigned the menu display with proper format and add spaces for clearer instructions after each command.
   f. History Display: I added a "Go Back" option in case user wanted to go back.
   g. QUIT message: I modified the quit method to log the [STOPPED] message before closing processes and added a small delay to make sure it went through.
   h. Case Handling: I converted all the inputs to be uppercase.
   i. Error Message: I make sure all error messages were properly captured and logged in the Logger.java file.
   j. Redundant History: I refined history to only keep unique entries.

k. Log File Format: I use the same format in the example showed in class to standardized log entry format to make sure all entries had proper timestamps and categories, i.e. [ENCRYPT], [DECRYPT], etc.

5. What did you learn doing this project?

   a. Planning was a really big thing for me, at first I just tried to go with the flow, it works for Logger and Encryption alone. But when moving on the Driver, there were lots of wrong format and errors I encounter. So I went through everything and make sure it works properly. I learn a bit on terminal design to make it easier for user to navigate between each command and ease of use. I also learned which data structure to use for specific use cases. I learned about inter-process communication in multi-process applications. I also learned repetitive is the key to programming, whether in testing the program or refinement to find which one works for me. I continuously fixing, testing and implementing new things to the program to make it works like the way I wanted. Overall, it was hard but it is worth it because I learned a lot.